

# Math 330 Abstract Algebra Notes

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## 1 Group Theory

### 1.1 Modular Arithmetic

**Definition 1.1.** Take  $n > 0$  and  $n \in \mathbb{Z}$ . We say that  $a, b \in \mathbb{Z}$  are *congruent modulo  $n$*  ( $a \equiv b \pmod{n}$ ) if  $n$  divides  $a - b$  ( $n|a - b$ ) or if the remainder of  $n/a$  equals the remainder of  $n/b$ .

Here are some properties:

1. if  $a \equiv b \pmod{n}$  and  $c \equiv d \pmod{n}$ , then  $a + c \equiv b + d \pmod{n}$ .

*Proof.*  $a \equiv b \pmod{n}$  implies that  $n|a - b$  which implies that  $\exists q \in \mathbb{Z}$  such that  $a - b = nq$ . The same can be said for  $a \equiv b \pmod{n}$  for some  $q' \in \mathbb{Z}$  such that  $c - d = nq'$ . Adding these equations together we have that  $a - b + c - d = (a + c) - (b + d) = n(q + q')$ . Since the integers are closed under addition,  $q + q'$  is an integer and therefore  $n|(a + c) - (b + d)$  and as a result,  $a + c \equiv b + d \pmod{n}$ .  $\square$

2. if  $a \equiv b \pmod{n}$  and  $c \equiv d \pmod{n}$ , then  $ac \equiv bd \pmod{n}$
3. Congruence mod  $n$  is an equivalence relation

### 1.2 Introduction to Groups

**Definition 1.2** (Group). A group is a set of elements  $G$  with a binary operation  $*$  that assigns to every pair  $(g, h)$  of elements of  $G$  another element in  $G$  denoted  $g * h$ . Groups follow the group axioms:

1. Associativity

$$(g_1 * g_2) * g_3 = g_1 * (g_2 * g_3)$$

2. There exists an element  $e \in G$  that is the identity such that

$$e * g_1 = g_1 * e = g_1$$

3. Every element has an inverse element denoted  $g^{-1}$  such that

$$g * g^{-1} = g^{-1} * g = e$$

4. The set of elements is closed under the group operation (although this is implied in the definition).

**Example 1.3.** Let us look at some examples

1. The set  $(\mathbb{Z}, +)$ . Addition is associative, the identity is 0, the inverse of each element  $g$  is  $-g$ .

2. The set  $(\mathbb{R}_{>0}, \cdot)$  where  $\cdot$  denotes regular multiplication. Regular multiplication is associative, the identity is 1, the inverse element is  $\frac{1}{g}$
3. The set  $(\{-2, -1, 0, 1, 2\}, +)$ . This fails due since it is not closed (take  $1 + 2 = 3 \notin$  the set
4. The set  $(\{\frac{1}{2^n}\}, \cdot)$ . This fails since there is no inverse.
5. Fix a positive integer  $n$ . Take the set  $(\{0, 1, 2, \dots, n-1\}, + \bmod n)$ . The identity is 0, addition is associative. The inverse of an element  $0 \leq a \leq n-1$  is  $n-a$  since  $(n-a) + a \bmod n = n \bmod n = 0$ . This group is denoted by Fraleigh as  $\mathbb{Z}$  but more generally denoted as  $\mathbb{Z}/n\mathbb{Z}$ .
6. Fix an  $n \in \mathbb{N}$ . We denote the group  $GL_n(\mathbb{R})$  as the set of invertible  $n \times n$  matrices with real entries with the operation of matrix multiplication. Matrix multiplication is associative, the identity is the  $n \times n$  identity matrix  $I_n$ . The inverse of every element  $A$  is the inverse matrix of  $A$ . This group is also known as the *General Linear Group*.